SIMATS SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105
TECHNICAL QUESTIONS FOR HBS SET -1

Engineer to Excel

Questions for First Round

1.
```
struct node
{
int i;
float j;
};
struct node *s[10];
```
The above C declaration defines

a. An array, each element of which is pointer to a structure of type node
b. A structure of 2 fields, each field being a pointer to an array of 10 elements
c. A structure of 3 fields: an integer, a float, and an array of 10 elements
d. An array, each element of which is a structure of type node

Answer (a)

2. The number of tokens in
```
printf("i = %d, &i – %x", i, &i);
```

a. 3
b. 10
c. 25
d. 22

Answer (b)

3. Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t, ignoring alignment
```
struct {
short s [5];
union {
float y;
long z;
}u;
} t;
```
a. 22 bytes
b. 18 bytes
c. 14 bytes
d. 10 bytes

Answer (b)

4. Consider the given three C functions:
```
[P1] int * g (void)
{
int x = 10;
return (&x);
}
```

```
[P2] int * g (void)
{
int * px;
*px = 10;
return px;
}
[P3] int * g (void)
{
int * px
px = (int *) malloc (sizeof(int));
*px = 10;
return px;
}
```

Which of the above three functions are likely to cause problems?

a. Only P1 and P2
b. Only P3
c. Only P1 and P3
d. P1, P2, and P3

Answer (a)

5. What does the given program print?

```
char c[ ] = “2011”
char *p = c;
printf (“%s”, p + p[3] – p[1]);
```

a. 2011
b. 2011
c. E2011
d. 011

Answer (b)

6. The output of the following C program is_____

```
void f1(int a, int b) {
int c;
c=a; a=b; b=c;
}
void f2(int *a, int *b) {
int c;
c=*a; *a=*b; *b=c;
}
int main(){
int a=4, b=5, c=6;
f1(a,b);
f2(&b, &c);
printf(“%d”,c-a-b);
}
```

a. -5
b. 6
c. -6
d. 0

Answer (a)

7. The following program prints _____

```
#include < stdio.h >
void f (int *p, int *q) {
p = q;
*p = 2;
}
int i = 0, j = 1;
int main ( ){
f(&i, &j);
printf (“%d %d \ n”, i, j);
return 0;
}
```

   a. 2 2
   b. 2 1
   c. 0 1
   d. 0 2

Answer (d)

8. Consider the following C program

```
void f(int, short);
void main()
{
int i = 100;
short s = 12;
short *p = &s;
_____ ; // call to f()
}
```

Which one of the following expressions, when placed in the blank above, will **NOT** result in a type checking error?

   a. f(s,*s)
   b. i = f(i,s)
   c. f(i,*s)
   d. f(i,*p)

Answer (d)

9. The output of the following C program is

```
#include< stdio.h >
struct Ournode{
char x,y,z;
};
int main(){
struct Ournode p = {‘1’, ‘0’, ‘a’+2};
struct Ournode *q = &p;
printf (“%c, %c”, *((char*)q+1), *((char*)q+2));
return 0;
```

   a. 0, c
   b. 0, a+2
   c. ‘0’, ‘a+2’
   d. ‘0’,’c’

Answer (a)

10. The output of the following C program is

```
#include < stdio.h >
```

```
void mystery(int *ptra, int *ptrb) {
int *temp;
temp = ptrb;
ptrb = ptra;
ptra = temp;
}
int main() {
int a=2016, b=0, c=4, d=42;
mystery(&a, &b);
if (a < c)
mystery(&c, &a);
mystery(&a, &d);
printf("%d\n", a);
}
```

a. 2016
b. 2018
c. 016
d. 16

Answer (a)

11. The output of the following C program is

```
#include < stdio.h >
int main () {
int arr [] = {1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip = arr+4;
printf ("%d\n", ip[1]);
return 0;
}
```

a. 6
b. 5
c. 66
d. 0

Answer (a)

12. The most appropriate matching for the following pairs

X: m=malloc(5); m= NULL;
Y: free(n); n->value = 5;
Z: char *p; *p='a';
1: using dangling
2: using uninitialized pointers
3. lost memory
is:

a. X-1 Y-3 Z-2
b. X-2 Y-1 Z-3
c. X-3 Y-2 Z-1
d. X-3 Y-1 Z-2

Answer (d)

13. Consider the following C function

```
void swap (int a, int b)
{
int temp;
temp = a;
```

```
a = b;
b = temp;
}
```

In order to exchange the values of two variables x and y

a. call swap (x,y)
b. call swap (&x, &y)
c. swap (x, y) cannot be used as it does return any value
d. swap (x,y) cannot be used as the parameters are passed by value

Answer (d)

14. Assume the following C variable declaration

```
int * A[10], B[10][10];
```

Of the following expressions

I. A[2]
II. A[2] [3]
III. B[1]
IV. B[2] [3]

Which will not give compile-time errors if used as left-hand sides of assignment statements in a C program?

a. I, II and IV
b. II, III and IV
c. II and IV
d. IV only

Answer (a)

15. In the C language

a) At most one activation record exists between the current activation record and the activation record for the main
b) The number of activation records between the current activation record and the activation record for the main depends on the actual function calling sequence.
c) The visibility of global variables depends on the actual function calling sequence.
d) Recursion requires the activation record for the recursive function to be saved on a different stack before the recursive function can be called.

a. There is no such restriction in C language
b. True
c. False. In C, variables are statically scoped, not dynamically
d. False. The activation records are stored on the same stack

Answer (b)

16. Consider the following C program:

```
#include <stdio.h>
int jumble(int x, int y){
x=2*x+y;
return x;
}
int main(){
int x=2, y=5;
y=jumble(y,x);
```

```
x=jumble(y,x);
printf("%d \n", x);
return 0;
}
```

The value printed by the program is _____

a. 26
b. 25
c. 20
d. 0

Answer (a)

17. Consider the following C program:

```
#include <stdio.h>
 int main(){
 int arr[]={1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4;
 printf("%d\n", ip[1]);
 return 0;
 }
```

The number that will be displayed on execution of the program is _____

a. 6
b. 7
c. 8
d. 0

Answer (a)

18. Consider the following C function.

```
void convert(int n){
if(n<0)
    printf("%d",n);
else {
        convert(n/2);
        printf("%d",n%2);
}
}
```

Which one of the following will happen when the function convert is called with any positive integer n as an argument?

a. It will print the binary representation of **n** and terminate
b. It will print the binary representation of **n** in the reverse order and terminate
c. It will print the binary representation of **n** but will not terminate
d. It will not print anything and will not terminate

Answer (d)

19. Consider the following C program:

```
#include <stdio.h>
int r(){
static int num=7;
return num–;
}
```

```
int main(){
for (r();r();r())
printf("%d",r());
return 0;
}
```

Which one of the following values will be displayed on execution of the programs?

a. 41
b. 52
c. 63
d. 630

Answer (b)

20. Consider the following C program:

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
j = j + j;
sum = sum + i/j;
printf("%f\n", sum);
}
return 0;
}
```

The number of times the variable sum will be printed, when the above program is executed, is _____

a. 0
b. 5
c. 1
d. None of the above

Answer (b)

CODING QUESTIONS FOR SECOND ROUND

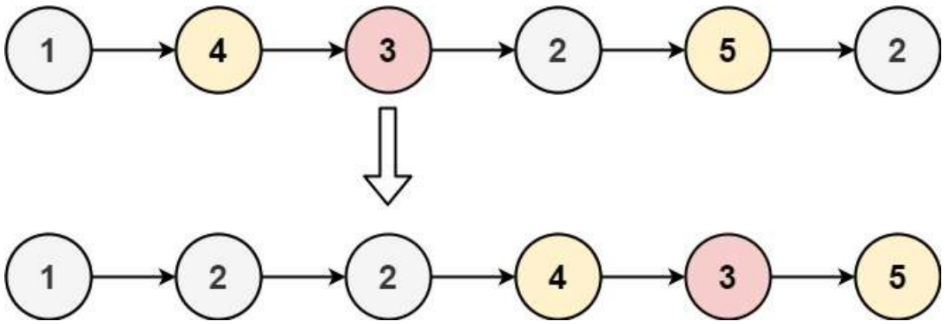
1.

Problem Statement

Given the head of a linked list and a value x, partition it such that all nodes less than x come before nodes greater than or equal x.

You should preserve the original relative order of the nodes in each of the two partitions.

Example 1



Input:
6
1 4 3 2 5 2
3

Output:
1->2->2->4->3->5

Example 2

Input:
2
2 1
2

Output:
1->2

2. Given a set of non-negative integers and a value **sum**, the task is to check if there is a subset of the given set whose sum is equal to the given **sum**.
   Examples:
   Input: set[] = {3, 34, 4, 12, 5, 2}, sum = 9
   Output: True
   Explanation: There is a subset (4, 5) with sum 9.
   Input: set[] = {3, 34, 4, 12, 5, 2}, sum = 30
   Output: False
   Explanation: No subset adds up to 30.

3. Find the shortest super sequence Z of given sequences X and Y such that both X and Y are sub-sequences of Z. For Example, consider the two following sequences, X and Y:
   X:ABCBDAB; Y: BDCABA

The length of the SCS is 9
The SCS are abcbdcaba, abdcabdab,and abdcbdaba

4. Find the longest sequence which can be obtained from the first original sequence by deleting some items and from the second original sequence by deleting other items.

X: ABCBDAB
Y: BDCABA

The length of the LCS is 4
LCS are BDAB, BCAB, and BCBA